

Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026

Tutorial-1 (Signal Energy/Power, RMS, KCL, KVL, Equivalent Resistance, Source Transformation)

The goal of this tutorial is to bridge the gap between theory and real-world application by solving practical signal and circuit problems using fundamental physical laws, fostering strong analytical intuition before introducing advanced systematic analysis.

Q1. You are designing a microcontroller circuit to heat a small $10\ \Omega$ resistive element using a 5 V source. The microcontroller outputs a 100 Hz PWM signal (0 V to 5 V pulses) to control the power.

A. Design the Duty Cycle: The heater requires an average power dissipation of exactly 0.5 Watts to reach the correct temperature. Calculate the required Duty Cycle (%) and the corresponding ON-time (T_{ON}) in milliseconds.

B. While the heater is running at this 0.5 W state, you connect a standard DC Voltmeter across it. What voltage value does the meter display? Can it be used to calculate the power supplied by the source?

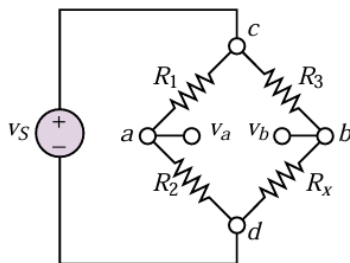
C. Calculate the RMS voltage of the signal you designed in Part A. Explain why this value is different from the voltmeter reading in Part B.

Q2. Lightning strikes or electrostatic discharges (ESD) are transient (non-periodic) events. These are modeled as energy signals rather than power signals because they do not exist forever. A lightning strike current is modeled as a decaying exponential: $i(t) = 20,000e^{-1000t}\text{ A}$ ($t \geq 0$), assume $i(t) = 0$ for $t < 0$.

A. Calculate the total Energy (E) dissipated if this current strikes a lightning rod with a grounding resistance of $10\ \Omega$.

B. Determine the total charge Q delivered by this surge.

Q3. A bridge circuit (as shown in the figure) is powered by a $V_s=10\text{ V}$ source.



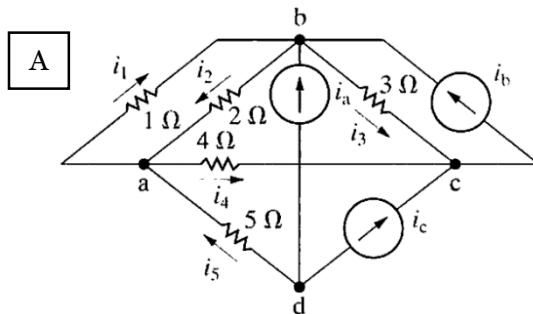
Resistor values: $R_1=1\text{ k}\Omega$, $R_2=2\text{ k}\Omega$, $R_3=5\text{ k}\Omega$.

Calculate R_x so that $V_{ab} = V_a - V_b$ is zero.

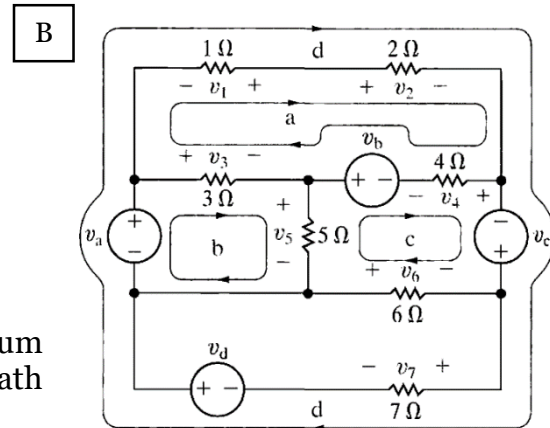
Q4. You have a solar array consisting of two cells connected in parallel. Cell A is modeled as a current source $I_A=2\text{ A}$ in parallel with internal resistance $R_A=10\Omega$. Similarly, Cell B is modeled as a current source $I_B=3\text{ A}$ in parallel with internal resistance $R_B=10\Omega$. The array is connected to 12 V, 20 Ah (Ampere-hour) Lead-Acid battery to charge it. The battery is modeled as a 12 V independent voltage source in series with an internal resistance 1.5Ω .

- Calculate the current flowing into the battery.
- What voltage would you measure at the battery terminals while it is charging?
- If the battery is initially completely empty (0% charge), how many hours will it take to fully charge it to 20 Ah? (Assume the current remains constant).

Q5. A. For circuit shown in A, write the sum of currents at each node. Note that there is no connection dot ($\bullet$) in the centre of the diagram, where the 4Ω branch crosses the branch containing the ideal current source i_a .

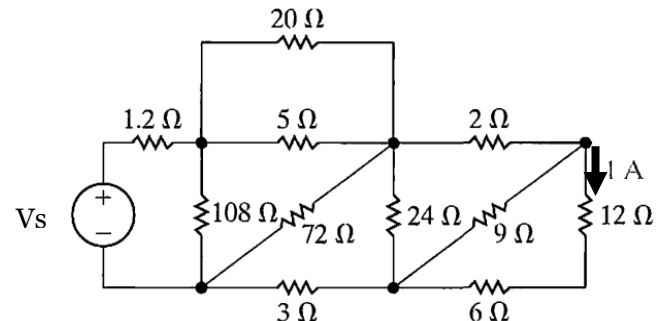


- For the circuit shown in B, write the sum of voltages around each designated path (i.e. path a, b, c and d).



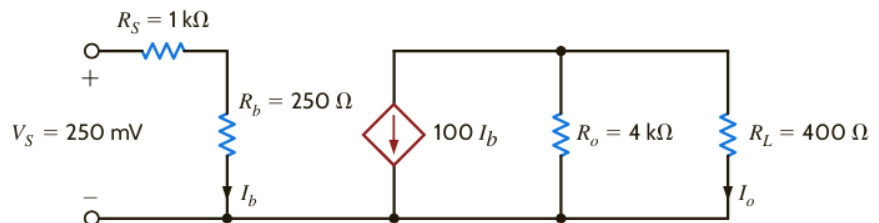
Q6. The current in the 12Ω resistor in the given circuit is 1 A.

- Using Kirchhoffs laws and Ohm's law, find V_s .
- Find the power dissipated in the 20Ω resistor.



Q7. Amplifiers (e.g. made from bipolar junction transistors) can be modeled as an equivalent circuit with current *dependent current sources* as shown in the figure.

Find the current through load R_L i.e. 400Ω resistor.



More Practice Problems

Practice questions in Chapter 1 and 2 problems in Irwin J.D. and Nelms R.M., Basic Engineering Circuit Analysis.

Q1. You want to help a friend jump-start her car with a dead battery using your car's battery. Your car's working battery has a voltage of 12.8 V. The friend's dead battery has a voltage of 11.0 V. You connect them with jumper cables that have a total series resistance of $0.15\ \Omega$.

- A. Draw the loop containing the two opposing voltage sources and the resistor.
- B. Calculate the initial surge current I flowing from the good battery to the dead battery.
- C. If you leave the cables connected for 5 minutes to charge the dead battery (assuming current stays roughly constant), calculate the total Charge (Q) transferred in Coulombs (C).
- D. Calculate the power being delivered *by* the good battery vs. the power being absorbed *by* the dead battery. Where does the missing power go?

Q2. *House wiring connects appliances in parallel. We use the fundamental laws in electrical circuits to check for safety.*

A kitchen circuit is protected by a 15 A circuit breaker. The voltage supply is 220 V. The following appliances are turned on simultaneously (connected in parallel):

- Toaster: Draws 1200 W.
 - Electric Kettle: Resistive element of $15\ \Omega$.
 - Microwave: Draws a current of 4 A.
- A. Calculate the current drawn by the Toaster.
 - B. Calculate the current drawn by the Kettle.
 - C. Find the total current I_{total} drawn from the source.
 - D. Will the 15 A circuit breaker trip?

Q3. A network consists of a voltage source $V_s=20\text{ V}$ in series with $R_s=5\ \Omega$. This combination is connected in parallel with another resistor $R_p=20\ \Omega$. This entire setup feeds a Load Resistor R_L .

If the Load Resistor $R_L=10\ \Omega$, calculate the current flowing through it using your simplified circuit.

[Hint: Convert the V_s, R_s series combination into a Current Source parallel with a resistor].

Q4. You are trying to power a space heater using a very long, cheap extension cord connected to a 220 V wall outlet.

- The Space Heater is modeled as a resistor $R_L = 10\ \Omega$.
 - The Extension Cord has a total wire resistance of $2\ \Omega$ (The current travels out on one wire and returns on the other, so assume the total loop resistance added by the wire is 2Ω).
- A. Draw the loop consisting of the 220 V source, the wire resistance (R_{wire}), and the heater load (R_L) all in series.
 - B. Calculate the current I flowing through the circuit.
 - C. Calculate the voltage actually appearing across the heater (V_L). Is it still 220 V?
 - D. Calculate the power dissipated as heat inside the extension cord.
 - E. If the extension cord is rated for a maximum of 50 Watts of heat dissipation before melting, is this setup safe?